# Letter of Motivation

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To whom it may concern,

During the third-year undergraduate course on Structural Analysis, I have been first introduced to finite element method. As part of the course we have developed a finite element program for the analysis of 2D frame structures using Mathcad. After I graduated, I have been always in touch with computational mechanics or programming.

I have graduated from COMMAS (Computational Mechanics of Materials and Structures) master program organized by Stuttgart University at 2003. I was the best student of my class. The title of my thesis was **"Comparison of alternative kinematical approaches to finite element formulations with embedded strong discontinuities"**. In this study I have implemented finite element formulations for elements cut by strong discontinuities, like cracks, based on elemental and nodal enrichments. I have developed a really original rederivation by myself to motivate locking free finite element formulation based on elemental enrichment, namely statically and kinematically optimal nonsymmetric (SKON) formulation. I have also implemented a damage type traction-separation constitutive law for tractions transmitted through cohesive macroscopic cracks based on internal variable formulation.

I also did my PhD study at Stuttgart University, Institute of Construction Materials, on a topic related with computational mechanics. The title of my thesis was **"High velocity impact and fragmentation of concrete: numerical simulation"**. During my PhD study, I have developed a finite element program, CIF (Contact Impact and Fragmentation), for 3D high velocity contact impact events from scratch. **Our simulations on perforation of circular plain-concrete slabs with a projectile was the only one in the literature which was able to show perforation limit documented in the experiments**. Within the context of high velocity contact-impact simulations, main ingredients of the finite element program are

- updated Lagrangian finite element formulation,
- explicit time integration,
- predictor-corrector type explicit/implicit contact algorithm,
- element deletion(erosion) algorithm,
- rate and temperature dependent microplane material model at large strains.

For this purpose, I have developed a program in the form of DLL to perform finite element computations using Intel Fortran. For pre and post processing tasks I have developed a GUI using Borland C++. I have also developed another program, CIF-JOB, basically to organize simulation projects and perform parametric studies using Borland C++.

I have worked at a software company, PROYA Ankara Turkey, for about ten years, 2008-2018, as a software engineer. I was the member of a software team gathered to develop COMOSYS (Construction Modelling Systems) program, an object-oriented software for the analysis, design and detailing of commercial and industrial buildings. In this program I was mainly responsible for the development of 3D rebar modelling module.

During the last year, 08.2018-07.2019, I have worked at Stuttgart University Institute of Construction

Materials as a research associate. It was a good opportunity for me to combine my past experiences from computational mechanics and programming in an optimal way. During this period, I have developed a really general-purpose finite element program for the simulation of different physical phenomena. Until I left, I have implemented simulation modules for heat transfer, solid mechanics, explicit dynamics and coupled thermo-mechanics. The program is highly modular and all modules,

- simulations,
- element formulations,
- material models,
- special algorithms (plugins), like contact, element erosion, explicit time integration,
- finite element utilities, like boundary conditions, time curves, initial conditions,

have been implemented in the form of DLLs. The interaction in between these different modules has been achieved by client-server type message passing interface. The above-mentioned structure has been constructed for both processor developed in Intel Fortran and pre-post processor developed in Borland C++.

I have also

- developed a finite element program, from scratch, using Matlab for the simulation of 3D large deformation frictionless contact problems together with mesh refinement/coarsening and remeshing algorithms.
- developed algortihms for 2D two-phase fluid flow simulations based on coupled Levelset-XFEM.
- developed an interface to Femap program for pre and post processing.
- implemented object rendering and object selection algorithms using OpenGL graphics library.

I have a very sound understanding of computational mechanics from both theoretical and implementation point of views. I have a dedicated and goal-oriented attitude. Within the light of my past experiences, as a student and as a professional, I am very competitive in all aspects related with computational mechanics and programming.

Sincerely yours,

Baris Irhan.